

Department of Geology and Geophysics

Subject name: Electrical methods of Prospecting (3-0-0) -3 Credit

Pre-requisite: None

Principle of electrical methods of prospecting, electrical properties and conduction in rocks, electrochemical properties. Electrical fields in geophysics, point source, line source, current and potential in the earth, Solution of Laplace's equation in Cartesian Coordinate, Cylindrical Polar Coordinate and Spherical Polar Coordinate, Equipotential and line of force (current flow), Concept of image in electrical methods. True and apparent resistivity, various electrode configuration and geometrical factors, investigation, exploration and penetration depth, S.P., I.P., and Charged body methods their principle, applications and interpretations, Theory of sounding over multi-layered earth, equivalence and suppression, resistivity transform and filter theory, Zar Zarrouk parameters, interpretation of VES curves, Theory of resistivity profiling and type curves. Principle of electrical resistivity imaging in 2-D and 3-D, modeling and interpretation.

Text Books

1. Applied Geophysics, Telford et al, 1990
2. Geosounding Principles I, Kefoed O, 1979.
3. Direct Current Electric sounding, Bhattacharya PK. And Patra, HP, 1968.

Reference books

1. Potential Theory and Applied Geophysics, K.K. Roy
2. An Introduction to Geophysical Exploration, P. Keary

S.No	Topic	Number of lectures
1	Principle of electrical methods of prospecting, electrical properties and conduction in rocks, electrochemical properties.	3
2	Electrical fields in geophysics, point source, line source, current and potential in the earth, Solution of Laplace's equation in Cartesian Coordinate, Cylindrical Polar Coordinate and Spherical Polar Coordinate, Equipotential and line of force (current flow),	3
3	Concept of image in electrical methods and electrical profiling	4
4	various electrode configuration and geometrical factors, investigation, exploration and penetration depth,	4
5	SP methods, data collection, interpretation	3
6	IP method, data collection, interpretation	3
7	Theory of sounding over multi-layered earth, equivalence and suppression, resistivity transform and filter theory,	5
8	Zar Zarrouk parameters, interpretation of VES curves, Theory of resistivity profiling and type curves.	3
9	Principle of electrical resistivity imaging in 2- and 3-D.	4
10	Case studies for various applications	4
	Total	36